

Multiple Choice

1. **Answer:** B

Options: A) Hair in the nose; B) Mucous membranes; C) Osteoblasts; D) Saliva

Note: Mucous membranes serve as a primary defense mechanism by trapping environmental bacteria and pathogens, preventing them from entering the body and thus playing a crucial role in the immune system.

2. **Answer:** A

Options: A) U.S.; B) Canadian; C) Swedish; D) French

Note: The U.S. has the highest teenage pregnancy rates compared to other developed countries due to a combination of factors including limited access to comprehensive sex education, insufficient contraceptive use, and socio-economic disparities.

3. **Answer:** A

Options: A) Prevalence; B) Incidence; C) Incipient; D) Diseased

Note: The correct answer is "Prevalence" because it measures the total number of individuals with a specific disorder in a population at a given time, calculated by dividing the number of affected individuals by the total population.

4. **Answer:** D

Options: A) hard palate.; B) hard palate and upper lip.; C) hard palate, upper lip and upper central incisor.; D) hard palate, upper lip, upper central incisor and lower first molar.

Note: The submandibular lymph nodes drain lymphatic fluid from the oral cavity, particularly from the hard palate, upper lip, upper central incisors, and lower first molars, making these areas potential sites for infection when lymphadenopathy is present.

5. **Answer:** A

Options: A) Texas; B) California; C) Hawaii; D) Vermont

Note: Texas is a more likely destination for older adults moving after retirement due to its favorable climate, lower cost of living, and availability of retirement communities and healthcare services.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Morning sickness typically occurs during the first trimester of pregnancy and usually subsides by the end of that period for most women, rather than continuing throughout the entire duration of pregnancy.

7. **Answer:** False

Options: A) True; B) False

Note: IUD use is linked to a variety of infections, including pelvic inflammatory disease and tubal infections, not just uterine infections, due to its potential effects on the reproductive system and the risk of introducing bacteria.

8. **Answer:** False

Options: A) True; B) False

Note: Other foramina, such as the infraorbital foramen and the greater palatine foramen, can also be palpated in a live patient, making the statement false.

9. **Answer:** False

Options: A) True; B) False

Note: The hypogastric region is located below the umbilical region in the lower abdomen, not just distal to the sternum, which is located in the upper thoracic region.

10. **Answer:** True

Options: A) True; B) False

Note: Hysterotomy involves surgical intervention similar to a cesarean section, making it more expensive and associated with higher risks of complications compared to other abortion methods.

Long Form Response

11. **Answer:** Developing an HIV vaccine presents significant challenges, particularly due to economic factors that hinder both production and accessibility. The complexity of the virus and the need for a robust immune response make research and development costly, often requiring substantial funding that may not be readily available. Additionally, the high costs associated with clinical trials and regulatory approval further inflate the expenses. Consequently, these financial barriers can lead to a situation where any potential vaccine is priced too high for widespread distribution, especially in low-resource settings where HIV is most prevalent. As a result, the combination of high production costs and limited funding creates a critical obstacle in making an effective HIV vaccine accessible to those who need it most.

Note: The answer correctly identifies that the high costs of research, development, clinical trials, and regulatory approvals, coupled with the complexity of the HIV virus, create significant economic barriers that impede both the production and accessibility of an effective vaccine.

12. **Answer:** Anxiety significantly impacts erectile function by disrupting the physiological processes necessary for sexual arousal, particularly through its effects on relaxation. Relaxation is crucial for achieving vasocongestion, which refers to the engorgement of blood vessels in the penis, a vital component of the erectile response. When an individual is anxious, heightened levels of stress hormones can inhibit the relaxation of smooth muscle and blood vessels, leading to insufficient blood flow and difficulty

achieving or maintaining an erection. Consequently, managing anxiety through relaxation techniques can enhance vasocongestion and promote a healthier sexual response, underscoring the interconnectedness of psychological well-being and physiological function in sexual health.

Note: correct as it clearly establishes the connection between anxiety and erectile dysfunction by explaining how anxiety disrupts relaxation, which is essential for vasocongestion and the overall physiological process of achieving an erection.

End of Answer Key